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## ► To cite this version:

Nadia Ben Hadj Boubaker, Nadia Yacoubi Ayadi, Zahra Kodia. Path-Based Explanations for Knowledge Graph-Driven Course Recommendation. Proceedings of the 31st Conference on Cooperative Information Systems (COOPIS'2025), Oct 2025, Marbella, Spain. <hal-05385586>

**HAL Id: hal-05385586**

**<https://hal.science/hal-05385586v1>**

Submitted on 27 Nov 2025

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# Path-Based Explanations for Knowledge Graph-Driven Course Recommendation

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**Abstract.** Recommendation systems (RS) play a key role in e-learning by guiding learners toward relevant educational resources. Yet, the integration of domain knowledge to enhance both accuracy and explainability remains underexplored. This paper presents an explainable e-learning RS grounded in an educational Knowledge Graph (KG). The KG is constructed by extracting and linking key course concepts, and leveraged through embedding techniques to improve recommendation quality. To ensure transparency, we propose a path-based explanation mechanism that identifies and ranks user–course connections using a scoring function combining similarity measures and random walk probabilities. A case study demonstrates that the approach not only improves recommendation accuracy but also generates diverse, interpretable explanations, contributing to more transparent and trustworthy systems.

**Keywords:** Recommendation Systems · Knowledge Graphs · Explainability · E-learning.

## 1 Introduction

Explainable Artificial Intelligence (XAI) has become essential need towards increasing trust and transparency in machine learning models, especially in domains where decision-making impacts users directly. Knowledge Graphs (KG) offer a promising solution by representing entities and their relationships in a structured, semantically rich form. The dual capability of supporting machine reasoning and enabling human-understandable explanations makes KG a natural fit for XAI [16]. In the context of recommender systems (RS), explainability can be achieved either through intrinsically transparent models or via post-hoc methods designed to justify system outputs [23]. Increasingly, KGs have been integrated into RS to address the limitations of conventional approaches by unifying preference propagation with KG embedding regularization [21, 6]. Explanations are typically generated by tracking semantic paths from user interactions to recommended items, thereby improving acceptance, satisfaction, and trust. Nevertheless, most existing approaches rely on open-domain KGs such as DBpedia or Freebase [3]. While these resources offer broad coverage, they often lack the domain-specific granularity required for precise and reliable explanations in specialized contexts such as education. To address this gap, we propose a post-hoc explainable framework for educational recommender

systems. In this paper, we introduce EduKA, a domain-specific knowledge graph that integrates user interactions with detailed metadata about learning materials. We compare multiple Knowledge Graph Embedding (KGE) methods to evaluate their impact on recommendation performance and present a case study illustrating how EduKA supports interpretable reasoning chains. Our findings demonstrate that domain-specific KGs can enhance both transparency and trustworthiness in educational recommendations.

The paper is structured as follows: Section 2 reviews related work on knowledge graph construction in e-learning contexts and their use in explainable recommender systems. In Section 3, we present our proposed approach, detailing the EduKA knowledge graph construction pipeline and comparing different knowledge graph embedding techniques to evaluate their effect on recommendation performance. Section 4 introduces a path-based post-hoc explainability method that leverages the structural richness of EduKA to generate interpretable recommendations. Finally, Section 5 summarizes our contributions and outlines our future work.

## 2 Related works

Recent advances in Machine Learning (ML) and Natural Language Processing (NLP) have stimulated growing interest in the automatic construction of KG [8]. In the educational domain, this task remains particularly challenging due to the difficulty of extracting core pedagogical elements such as learning outcomes, prerequisite relations, and domain-specific concepts [1]. To address these challenges, Lu et al. [12] introduced DS-MOCE, which combines pre-trained language models with domain-specific embeddings to extract course concepts from MOOCs, thereby mitigating issues of noisy or incomplete annotations. Similarly, Reales et al. [17] leveraged large language models (LLMs) to construct semantic KGs using pipelines that integrate zero-shot triple extraction with DBpedia-based enrichment.

Beyond their construction, KG have also been increasingly applied in e-learning recommender systems to improve both personalization and interpretability [7]. Ain et al. [2] modeled learner knowledge through personal KGs enriched with SBERT embeddings, while Boubaker et al. [4] proposed KA-ERN, which combines KG embeddings with ensemble deep learning for personalized recommendations. Troussas et al. [20] designed a path-based system linking students and activities through relations such as difficulty or cognitive level, with recommendations generated via meta-paths and pattern mining. While these approaches demonstrate high recommendation accuracy, they often lack explainability, which remains essential for fostering user trust and understanding.

To address this limitation, post-hoc explanation techniques such as LIME [18] and SHAP [13] have been applied to clarify recommendation processes. Nevertheless, these approaches often struggle to provide explanations that are both intuitive and aligned with domain knowledge, particularly when neural networks are employed in complex domains such as healthcare [23] and education [15]. In this context, our work contributes to the growing research on explainable educational recommender systems. We introduce EduKA, a purpose-built educational knowledge graph that captures rich semantic relationships among users, courses, and domain concepts. Leveraging EduKA, we generate post-hoc explanations in the form of ranked paths, enabling learners to trace and better comprehend the reasoning behind recommendations.

### 3 Proposed approach

In this section, we introduce the pipeline of the proposed approach, which comprises three main phases: (1) the EduKA (Educational Knowledge GrAph) modeling and construction, (2) the recommendation generation, and (3) the explainability generation. Figure 1 illustrates the overall workflow of the proposed approach.

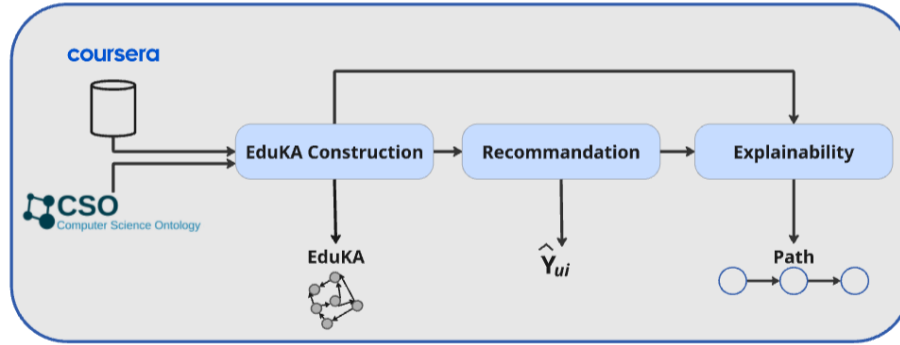


Fig. 1: The workflow of our approach

#### 3.1 EduKA graph modeling and construction

The EduKA knowledge graph<sup>3</sup> has been developed as part of this research to support explainable recommendations in educational contexts. It is modeled as an RDF graph that integrates both user-item interaction data and structured domain knowledge considering a Coursera dataset<sup>4</sup> (Figure 2).

The graph construction process is based on a multi-stage pipeline that combines state-of-the-art Natural Language Processing (NLP) and knowledge modeling. It includes automated concept extraction, entity linking, and semantic graph construction, enabling the transformation of unstructured textual data into a structured knowledge representation. The key components of this process are described in detail in following.

**Course Concept (CC) Extraction** To identify course-related concepts, we adopt a dual strategy leveraging two complementary methods. First, we apply a keyphrase extraction approach based on the KBIR framework [9], which combines contextual semantic understanding and candidate ranking. The model, built upon a RoBERTa encoder [11] and trained on the Inspec benchmark, achieves state-of-the-art performance in extracting domain-relevant keyphrases. This makes it particularly suitable for capturing the core educational topics expressed in course descriptions.

<sup>3</sup> The RDF version of EduKA is available at <https://zenodo.org/records/15804872>

<sup>4</sup> <https://www.kaggle.com/datasets/khusheekapoor/coursera-courses-dataset-2021>

In parallel, we employ the DBpedia Spotlight API <sup>5</sup> to detect and link named entities present in both course titles and descriptions. A confidence threshold of 0.8 is applied to ensure precision and contextual relevance in the entity recognition process. This ensures that a subset of extracted entities are reliably aligned with concepts in the DBpedia knowledge base, thereby facilitating semantic grounding within a structured RDF graph.

**Entity disambiguation and Linking** Entity linking (EL) involves the task of mapping course keyphrases extracted in the first step of the pipeline to structured domain knowledge formalized in domain ontologies. Specifically, we leverage the Computer Science Ontology (CSO) <sup>6</sup> to support domain-specific entity linking, as the majority of courses in our dataset fall within the computer science field. In order to disambiguate the outputs of CC extraction step, we go through a two-fold process:

- **Ontology-based Entity Linking:** The CSO provides a comprehensive and hierarchically structured vocabulary for computer science topics. To operationalize its use, we transform the ontology into a dictionary-based representation, enabling efficient matching and grouping of semantically equivalent or related terms under unified labels. Additionally, a curated subset of the CSO was developed to reduce redundancy and ensure consistency in concept mapping across all educational resources within the graph.
- **Fuzzy String for Approximate matching:** In cases where exact string matching with CSO concepts fail, we integrate the FuzzyWuzzy library <sup>7</sup> which performs approximate string matching using the Levenshtein distance metric. It is particularly well-suited for identifying approximate matches and effectively handling minor variations in word order, spelling, and structure. In our work, FuzzyWuzzy was integrated to handle cases where course keywords did not directly match any concept in the CSO or Basic Concepts List. By applying a high similarity threshold of 0.9, we ensured that only relevant and semantically meaningful matches were accepted, reducing false positives.

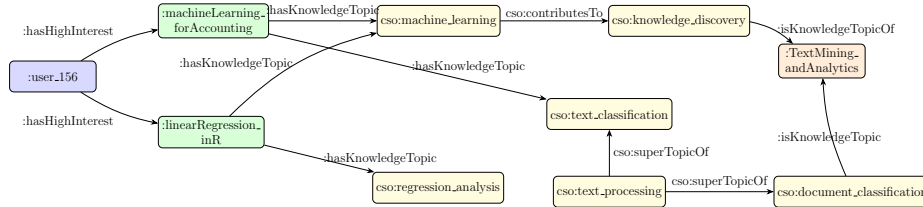


Fig. 2: An EduKA sub-graph illustrating user-item interactions, CSO course concepts and their inter-links.

**KG Modeling and Generation** The RDFLib Python library is used to construct the RDF graph, structuring course metadata (e.g., title, difficulty level) according to open vocabularies such as Schema.org <sup>8</sup> and Dublin Core standards. In EduKA,

<sup>5</sup> <https://www.dbpedia-spotlight.org/api>

<sup>6</sup> <https://cso.kmi.open.ac.uk/home>

<sup>7</sup> <https://pypi.org/project/fuzzywuzzy/>

<sup>8</sup> <https://schema.org/>

each course is semantically linked to CSO topics using the *eduka:hasKnowledgeTopic* property. Additionally, user-item interactions are integrated into the graph, categorized into three levels—*eduka:smallInterest*, *eduka:mediumInterest*, and *eduka:highInterest*—to capture varying user engagement levels with courses. Table 1 provides an overview of the statistics of the EduKA knowledge graph, including user-item interactions and the structural properties of the underlying knowledge graph.

Table 1: Graph Statistics

|                       |                           |
|-----------------------|---------------------------|
| User-Item dataset     | # of Users: 1,000         |
|                       | # of Items: 3,533         |
|                       | # of Interactions: 68,801 |
| EduKA Knowledge graph | # of Entities: 15,300     |
|                       | # of Entity Types: 3      |
|                       | # of Relation Types: 11   |
|                       | # of Triplets: 111,304    |

### 3.2 Recommendation Generation

In this section, we present a comprehensive evaluation designed to assess the performance of several KGE techniques in the context of recommendation systems. The embeddings generated by different approaches were used as input features for multiple classification models, enabling us to evaluate their predictive capacity in identifying relevant user-item interactions. The goal of this phase is not to introduce a new recommendation model, but rather to assess the suitability of various knowledge graph embedding techniques in supporting effective recommendations. We explore how different embeddings capture well the relational and semantic structure of the graph and support the recommendation task.

Thus, we have investigated five embedding methods that represent diverse approaches to graph representation learning: DistMult [22], TransR [10], Node2Vec [5], DeepWalk [14], and RDF2Vec [19]. These techniques were selected to cover both translational distance models (e.g., TransR), bilinear semantic models (e.g., DistMult), and path-based or random-walk strategies (e.g., Node2Vec, DeepWalk, RDF2Vec), thereby enabling a comparative analysis from topological and semantic perspectives.

Each embedding method was first used to generate vector representations for all entities in the EduKA knowledge graph. For every user-course pair, the embeddings of the user and the course were concatenated into a single feature vector  $[f(u)||f(i)]$ , capturing both user interests and course characteristics. These vectors were then used as input to three classifiers: Multi-Layer Perceptron (MLP), Random Forest (RF), and XGBoost in order to predict the relevance of each user-course interaction. The resulting predictions were treated as recommendations, and model performance was evaluated using Precision, Recall, and F1-score metrics providing a quantitative evaluation of how well the embeddings supported accurate recommendations.

As presented in Table 2, RDF2Vec and TransR combined with MLP deliver the best overall performance, with F1-scores of 0.985 and 0.926, respectively, indicating an excellent balance between precision and recall. DistMult performs well only when paired

Table 2: Recommendation experiments: performance of different embedding methods and classifiers.

| Embedding | Classifier | Precision    | Recall       | F1           |
|-----------|------------|--------------|--------------|--------------|
| DistMult  | MLP        | 0.420        | 0.322        | 0.364        |
|           | RF         | 0.436        | <b>0.990</b> | 0.131        |
|           | XGBoost    | <b>0.943</b> | 0.986        | <b>0.964</b> |
| Node2Vec  | MLP        | 0.828        | <b>0.815</b> | <b>0.821</b> |
|           | RF         | <b>0.945</b> | 0.412        | 0.574        |
|           | XGBoost    | 0.839        | 0.669        | 0.745        |
| DeepWalk  | MLP        | 0.750        | <b>0.857</b> | 0.800        |
|           | RF         | 0.714        | 0.833        | <b>0.769</b> |
|           | XGBoost    | <b>0.944</b> | 0.586        | 0.723        |
| RDF2Vec   | MLP        | 0.771        | 0.900        | <b>0.985</b> |
|           | RF         | 0.769        | 0.269        | 0.399        |
|           | XGBoost    | <b>0.846</b> | <b>0.919</b> | 0.881        |
| TransR    | MLP        | <b>0.908</b> | <b>0.944</b> | <b>0.926</b> |
|           | RF         | 0.836        | 0.880        | 0.834        |
|           | XGBoost    | 0.876        | 0.880        | 0.878        |

with XGBoost but shows very low performance with other classifiers, highlighting its strong dependence on the classification model. Random-walk-based methods, such as Node2Vec and DeepWalk, achieve intermediate performance. Among the classifiers, MLP consistently yields the best results, while RF favors high recall at the expense of precision, leading to lower F1-scores.

These findings contribute directly to the selection of embedding techniques for the KG-based recommendation framework. They demonstrate that certain models, particularly TransR and RDF2Vec, support accurate classification. In section 4, we demonstrate how they also preserve semantic signals that can support post-hoc interpretability.

## 4 Explainability Generation

While KGE-based methods capture relational patterns effectively, their latent representations remain opaque, limiting transparency in recommendation predictions. In educational settings, where trust and accountability are critical, explainability is essential to reveal which semantic and structural elements of the graph drive user-course recommendations. To address this, we introduce an explanation approach that exploits the semantic and structural richness of the EduKA knowledge graph. Our method generates path-level post-hoc explanations by exploring all user-item paths and presenting them as interpretable evidence chains. These chains leverage the explicitly defined semantics of nodes (e.g., topics, sub-topics, related topics) and edges (e.g., hasInterestIn, isKnowledgeTopicOf) to provide human-understandable justifications for recommendations. The underlying intuition is that considering multiple alternative paths highlights the diversity and robustness of the recommendation, showing not only why a course is relevant from one perspective but also reinforcing its validity through different semantic connections. Therefore, we propose a Path Ranking Approach (PRA), which systematically scores

and ranks candidate paths based on their semantic coherence and structural plausibility. As illustrated in Table 3, the candidate paths may vary considerably in terms of length, informativeness, and clarity. Through a case study analysis, we show that PRA enables the selection of the most salient explanation paths, ensuring both relevance and interpretability in the context of user-centered educational recommendations.

#### 4.1 Path Ranking approach

Given a knowledge graph be denoted by  $\mathcal{G} = (\mathcal{E}, \mathcal{R})$ , where  $\mathcal{E}$  is the set of entities (e.g., learners, courses, concepts) and  $\mathcal{R}$  is the set of directed relations (e.g., `eduka:highInterest`, `eduka:hasKnowledgeTopic`, `cso:ContributesTo`). Let a set of paths  $\mathcal{P} = \{p_1, p_2, \dots, p_N\}$  such as each path  $p_i$  is defined as follows:

$$p_i = (v_0 = u, v_1, v_2, \dots, v_L = i)$$

connecting the learner node  $u$  to a recommended course  $i$ . We assign to each path  $p_i$  a ranking score  $S(p_i)$  defined as:

$$S(p_i) = \lambda \cdot \tilde{S}_{\text{jac}}(p_i) + (1 - \lambda) \cdot \tilde{S}_{\text{rw}}(p_i) \quad (1)$$

where:

- $\tilde{S}_{\text{jac}}(p_i)$  is the normalized semantic similarity score based on jaccard index,
- $\tilde{S}_{\text{rw}}(p_i)$  is the normalized relevance score based on random walk probability,
- $\alpha, \beta \in [0, 1]$  are weighting parameters satisfying  $\alpha + \beta = 1$ , controlling the influence of each component.

**Path similarity via jaccard index** Each path  $p_i$  is first transformed into a token set  $T(p_i)$  containing the normalized labels of all nodes and edges along the path. This representation captures the key semantic and ontological elements involved in the explanation.

$$S_{\text{jac}}(p_i) = \frac{1}{N} \sum_{j=1}^N \frac{|T(p_i) \cap T(p_j)|}{|T(p_i) \cup T(p_j)|} \quad (2)$$

This formulation favors paths that exhibit greater lexical overlap with others, highlighting recurring semantic patterns in the explanation space. For comparability across paths, the raw Jaccard scores are normalized via min-max scaling:

$$\tilde{S}_{\text{jac}}(p_i) = \frac{S_{\text{jac}}(p_i) - \min_k S_{\text{jac}}(p_k)}{\max_k S_{\text{jac}}(p_k) - \min_k S_{\text{jac}}(p_k)} \quad (3)$$

**Random walk-based path scoring** To capture both structural accessibility and semantic relevance of path-level explanations, we propose an improved scoring mechanism that integrates transition probabilities, relation semantics, and length normalization. The score of the path  $p_i$  is calculated as:

$$S_{\text{rw}}(p_i) = \left( \prod_{\ell=0}^{L-1} \mathbb{P}(v_{\ell+1} | v_\ell) \cdot w(e_{\ell, \ell+1}) \right)^{\frac{1}{L}} \quad (4)$$

where:

- $\mathbb{P}(v_{\ell+1}|v_\ell)$  is the transition probability from node  $v_\ell$  to  $v_{\ell+1}$ , typically defined as:

$$\mathbb{P}(v_{\ell+1}|v_\ell) = \frac{1}{\deg^+(v_\ell)}$$

with  $\deg^+(v_\ell)$  being the out-degree of node  $v_\ell$ .

- $w(e_{\ell,\ell+1}) \in \mathbb{R}^+$  is an edge importance weight reflecting the semantic contribution of relation  $e_{\ell,\ell+1}$  to the path's value.
- The exponent  $\frac{1}{L}$  ensures geometric mean normalization, mitigating the penalization of longer paths.

After computing  $S_{\text{rw}}(p_i)$  for all candidate paths, scores are normalized using min-max scaling:

$$\tilde{S}_{\text{rw}}(p_i) = \frac{S_{\text{rw}}(p_i) - \min_k S_{\text{rw}}(p_k)}{\max_k S_{\text{rw}}(p_k) - \min_k S_{\text{rw}}(p_k)} \quad (5)$$

The final path ranking score is a weighted combination of these two components, balancing semantic coherence with graph connectivity. To confirm the effectiveness and interpretability of this approach, we present a case study that illustrates how highly ranked paths are both meaningful and informative, providing transparent, interpretable explanations for recommended courses while emphasizing patterns common across learners' interactions.

## 4.2 Case study analysis

In order to evaluate the PRA approach, we design a case study that illustrates its effectiveness in providing and ranking explanation paths. Specifically, we consider a scenario where the course *Text Mining and Analytics* is recommended to a given user ( $\text{user}_{156}$ ). For this user-item pair, we systematically extract all possible paths within the EduKA knowledge graph that connect the user to the recommended course.

In total, we identified 10 distinct paths and grouped them according to their structural patterns. Four main path patterns were distinguished, as illustrated in Table 3. The reported frequency indicates how many explanation paths of a given pattern were found in our running example. These patterns highlight both the diversity and richness of the explanations, as well as the system's ability to uncover various meaningful connections that justify the recommendation. Each pattern reflects a specific way in which the user's prior learning history is semantically connected to the recommended course.

Table 3: Frequent explanation path patterns observed in the knowledge graph

| Path pattern                                                                                                                                                                    | Frequency |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| $u \xrightarrow{\text{hasInterest}} c \xrightarrow{\text{hasKnowledgeTopic}} t_1 \xrightarrow{\text{contributesTo}} t_2 \xrightarrow{\text{isKnowledgeTopicOf}} c_{\text{rec}}$ | 5         |
| $u \xrightarrow{\text{hasInterest}} c \xrightarrow{\text{hasKnowledgeTopic}} t \xrightarrow{\text{isKnowledgeTopicOf}} c_{\text{rec}}$                                          | 3         |
| $u \xrightarrow{\text{hasInterest}} c \xrightarrow{\text{hasKnowledgeTopic}} t_1 \xrightarrow{\text{superTopicOf}} t_2 \xrightarrow{\text{isKnowledgeTopicOf}} c_{\text{rec}}$  | 1         |
| $u \xrightarrow{\text{hasInterest}} c \xrightarrow{\text{hasKnowledgeTopic}} t_1 \xrightarrow{\text{superTopicOf}^{-1}} t_2 \xrightarrow{\text{superTopicOf}} c_{\text{rec}}$   | 1         |
| $t_3 \xrightarrow{\text{isKnowledgeTopicOf}} c_{\text{rec}}$                                                                                                                    |           |

Once the candidate paths are extracted, we apply the Path Ranking Algorithm (PRA) to assign scores and rank them. Table 4 presents the results, showing the normalized Jaccard similarity, the normalized random walk probability, and the final aggregated score computed using a weighted combination of both values. In our experiments, we set  $\alpha=0.3$  and  $\beta=0.7$ , assigning more importance to Jaccard similarity. This configuration yielded the best performance.

Table 4: List of paths and their corresponding scores ( $\alpha=0.3$ ,  $\beta=0.7$ )

| Path ID  | Nor. Jac. Sim. | Nor. RW Prob. | Final Score |
|----------|----------------|---------------|-------------|
| $p_2$    | 0.589          | 0.472         | 0.804       |
| $p_3$    | 0.558          | 0.562         | 0.772       |
| $p_1$    | 0.573          | 0.472         | 0.740       |
| $p_9$    | 0.545          | 0.562         | 0.720       |
| $p_5$    | 0.563          | 0.368         | 0.596       |
| $p_4$    | 0.516          | 0.464         | 0.508       |
| $p_7$    | 0.512          | 0.472         | 0.501       |
| $p_6$    | 0.468          | 0.562         | 0.418       |
| $p_8$    | 0.411          | 0.668         | 0.300       |
| $p_{10}$ | 0.413          | 0.584         | 0.225       |

According to this ranking, the top-scored paths were  $P_2$ ,  $P_3$ , and  $P_1$ . Their detailed structures are shown below:

```
P1 : user_156
    --[hasHighInterest]--> Machine Learning for Accounting
    --[hasKnowledgeTopic]--> data_analytics
    --[ContributesTo]--> knowledge_discovery
    --[isKnowledgeTopicOf]--> Text Mining and Analytics
```

```
P2: user_156
    --[hasHighInterest]--> Machine Learning for Accounting
    --[hasKnowledgeTopic]--> machine-learning
    --[ContributesTo]--> knowledge_discovery
    --[isKnowledgeTopicOf]--> Text Mining and Analytics
```

```
P3: user_156 --[
    hasHighInterest]--> Building and analyzing LR model in R
    --[hasKnowledgeTopic]--> machine-learning
    --[ContributesTo]--> knowledge_discovery
    --[isKnowledgeTopicOf]--> Text Mining and Analytics
```

Overall, this case study underscores several strengths of the proposed approach. First, it enables multi-hop reasoning, allowing the system to move beyond direct user-item interactions and uncover deeper semantic connections across multiple levels of the knowledge graph (Table 3). Second, the diversity of the extracted paths enriches the explanations and ensures they capture multiple facets of user interests. In summary, the

qualitative analysis demonstrates that our method produces human-understandable, diverse, and meaningful explanations by effectively leveraging the structural and semantic richness of the educational knowledge graph.

## 5 Conclusions and Future Works

In this paper, we introduced an explainable recommender system specifically designed for the e-learning domain. Our work explored the potential of leveraging an educational knowledge graph, EduKA, to enhance both the accuracy of course recommendations and the transparency of their underlying rationale. By integrating knowledge graph embeddings with a path-ranking explanation mechanism, we demonstrated how recommendation accuracy can be improved while also providing diverse and semantically meaningful justifications. Beyond the technical contributions, this work highlights the importance of explainability in fostering trust and adoption of recommender systems in education. Explanations that are grounded in domain concepts not only help learners understand why a course is suggested but also support metacognitive processes such as reflecting on one's prior knowledge and planning learning trajectories.

As future work, we aim to conduct a comprehensive user study to evaluate the usefulness, clarity, and perceived fairness of the generated explanations. Such a study will provide insights into how learners with different backgrounds and goals interpret the explanations, and whether these explanations effectively support decision-making in practice. Ultimately, this research contributes to the growing field of explainable educational recommender systems and provides a groundwork for building user-centric, transparent, and pedagogically aligned recommendation technologies.

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